LESSON PLAN

MOCKUP SQUADRON ENVIRONMENT (MSE) (24)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	16
Total No. of Lesson Plans	21	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Demonstration on insertion and extraction of pins of aircraft electrical plugs.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Demonstration on insertion and extraction of pins of aircraft electrical plugs.
5. Specific Objectives	24.13.1.1 Trainee will be able to carry out workshop maintenance on aircraft cables and PCBs.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration on insertion and extraction of pins of aircraft electrical plugs; Safety precautions: Wear gloves and eye protection; Instructor supervision: Direct observation and correction of technique.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of aircraft electrical plugs?
(b) Why is proper pin alignment critical during insertion?
(c) What safety gear is required before handling aircraft electrical connectors?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Identification of pin types and configurations • Pre-insertion inspection and cleaning procedures • Correct alignment and insertion technique • Verification of secure locking mechanism • Proper extraction and removal sequence	L & Ds	PPT, IB	15 min
(b)	• Common errors in pin insertion and extraction • Troubleshooting damaged or corroded pins • Handling of different voltage ratings safely • Documentation of maintenance actions • Storage and protection of removed plugs	L & Ds	PPT, PDF	20 min
(c)	• Final verification of plug integrity • Reassembly and testing procedures • Reporting discrepancies to maintenance control • Review of safety protocols • Preparation for periodic inspection	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Describe the correct sequence for extracting pins from an aircraft electrical plug.
(b) Explain the potential consequences of forcing pins into a misaligned connector.
(c) How do you verify that a plug is securely locked after insertion?

12. Home Assignment and Follow Up	Prepare a brief report on the differences between standard and high-voltage aircraft electrical plugs, referencing the procedures covered in LP 02.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Explanation on the evaluation of periodicity and level of calibration of STTE and CRETE.	Refer to the master syllabus reference list	LP 17

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min